

Chapter 11: Derivatives of Multivariable Functions (§11.1 - 11.4)

Instructor: Shengwen Guo

Department of Mathematics, St. Bonaventure Univ.

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Review of the Definite Integral

Let's recall the basic facts concerning definite integrals of functions of a single variable.

- $f(x)$ is defined on $[a, b]$.
- we start by dividing $[a, b]$ into n subintervals $[x_{i-1}, x_i]$ of equal width $\Delta x = (b - a)/n$ and choose $x_i^* \in [x_{i-1}, x_i]$.

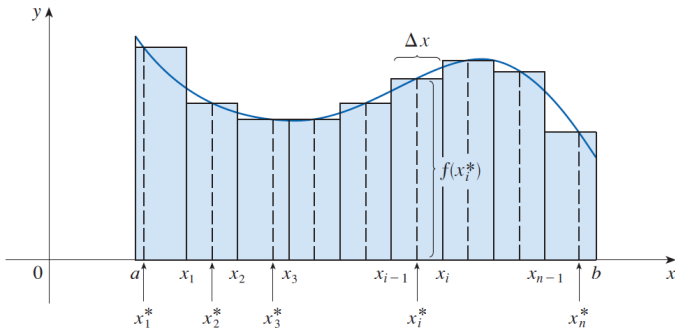
Then we form the **Riemann sum**

$$\sum_{i=1}^n f(x_i^*) \Delta x$$

The definite integral of f from a to b is given by

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

In the special case where $f(x) \geq 0$, the Riemann sum can be interpreted as the sum of the areas of the approximating rectangles, and $\int_a^b f(x)dx$ represents the area under the curve $y = f(x)$ from a to b .



§11.1 Double Riemann Sums and Double Integrals over Rectangles

In our upcoming study of integral calculus for multivariable functions, we will first extend the idea of the single-variable definite integral to functions of two variables over rectangular domains. To do so, we will need to understand how to partition a rectangle into subrectangles. Let R be rectangular domain $R = \{(x, y) : 0 \leq x \leq 6, 2 \leq y \leq 4\}$ (we can also represent this domain with the notation $[0, 6] \times [2, 4]$), as pictured in [Figure 11.1.1](#).

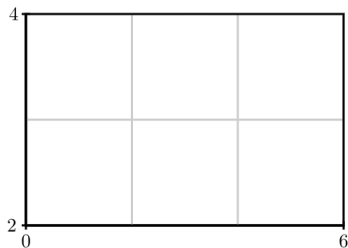


Figure 11.1.1. Rectangular domain R with subrectangles.

To form a partition of the full rectangular region, R , we will partition both intervals $[0, 6]$ and $[2, 4]$; in particular, we choose to partition the interval $[0, 6]$ into three uniformly sized subintervals and the interval $[2, 4]$ into two evenly sized subintervals as shown in [Figure 11.1.1](#). In the following questions, we discuss how to identify the endpoints of each subinterval and the resulting subrectangles.

- i. Let $0 = x_0 < x_1 < x_2 < x_3 = 6$ be the endpoints of the subintervals of $[0, 6]$ after partitioning. What is the length Δx of each subinterval $[x_{i-1}, x_i]$ for i from 1 to 3?
- ii. Explicitly identify x_0 , x_1 , x_2 , and x_3 . On [Figure 11.1.1](#) or your own version of the diagram, label these endpoints.
- iii. Let $2 = y_0 < y_1 < y_2 = 4$ be the endpoints of the subintervals of $[2, 4]$ after partitioning. What is the length Δy of each subinterval $[y_{j-1}, y_j]$ for j from 1 to 2? Identify y_0 , y_1 , and y_2 and label these endpoints on [Figure 11.1.1](#).
- iv. Let R_{ij} denote the subrectangle $[x_{i-1}, x_i] \times [y_{j-1}, y_j]$. Appropriately label each subrectangle in your drawing of [Figure 11.1.1](#). How does the total number of subrectangles depend on the partitions of the intervals $[0, 6]$ and $[2, 4]$?
- v. What is area ΔA of each subrectangle?

Double Riemann Sums over Rectangles

In multivariable calculus, we will eventually develop the idea of a definite integral over a closed, bounded region (such as the interior of a circle). We begin with a simpler situation by thinking only about rectangular domains. Suppose that

- $f = f(x, y)$ is a continuous function defined on a rectangular domain

$$R = \{(x, y) : a \leq x \leq b, c \leq y \leq d\}$$

- We want to partition R into subrectangles. We do this by partitioning each of the intervals $[a, b]$ and $[c, d]$ into subintervals and using those subintervals to create a partition of R into subrectangles.

Activity 11.1.2. Let $f(x, y) = 100 - x^2 - y^2$ be defined on the rectangular domain $R = [a, b] \times [c, d]$. Partition the interval $[a, b]$ into four uniformly sized subintervals and the interval $[c, d]$ into three evenly sized subintervals as shown in [Figure 11.1.2](#). As we did in [Preview Activity 11.1.1](#), we will need a method for identifying the endpoints of each subinterval and the resulting subrectangles.

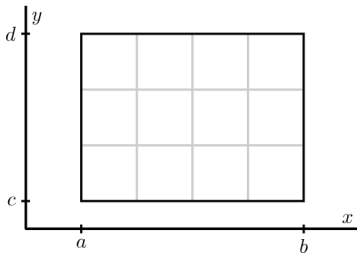


Figure 11.1.2. Rectangular domain with subrectangles.

- a. Let $a = x_0 < x_1 < x_2 < x_3 < x_4 = b$ be the endpoints of the subintervals of $[a, b]$ after partitioning. Label these endpoints in [Figure 11.1.2](#).
- b. What is the length Δx of each subinterval $[x_{i-1}, x_i]$? Your answer should be in terms of a and b .
- c. Let $c = y_0 < y_1 < y_2 < y_3 = d$ be the endpoints of the subintervals of $[c, d]$ after partitioning. Label these endpoints in [Figure 11.1.2](#).
- d. What is the length Δy of each subinterval $[y_{j-1}, y_j]$? Your answer should be in terms of c and d .
- e. The partitions of the intervals $[a, b]$ and $[c, d]$ partition the rectangle R into subrectangles. How many subrectangles are there?
- f. Let R_{ij} denote the subrectangle $[x_{i-1}, x_i] \times [y_{j-1}, y_j]$. Label each subrectangle in [Figure 11.1.2](#).
- g. What is area ΔA of each subrectangle?

- h. Now let $[a, b] = [0, 8]$ and $[c, d] = [2, 6]$. Let (x_{11}^*, y_{11}^*) be the point in the upper right corner of the subrectangle R_{11} . Identify and correctly label this point in [Figure 11.1.2](#). Calculate the product

$$f(x_{11}^*, y_{11}^*)\Delta A.$$

Explain, geometrically, what this product represents.

- i. For each i and j , choose a point (x_{ij}^*, y_{ij}^*) in the subrectangle $R_{i,j}$. Identify and correctly label these points in [Figure 11.1.2](#). Explain what the product

$$f(x_{ij}^*, y_{ij}^*)\Delta A$$

represents.

- j. If we were to add all the values $f(x_{ij}^*, y_{ij}^*)\Delta A$ for each i and j , what does the resulting number approximate about the surface defined by f on the domain R ? (You don't actually need to add these values.)
- k. Write a double sum using summation notation that expresses the arbitrary sum from part (j).

Volumes and Double integrals

Suppose that

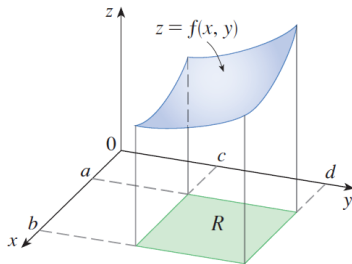
- $f = f(x, y) \geq 0$ and is defined on

$$R = [a, b] \times [c, d] = \{(x, y) \in \mathbb{R}^2 : a \leq x \leq b, c \leq y \leq d\}.$$

The graph of f is a surface given by $z = f(x, y)$.

- Let S be the solid that lies above R and under the graph of f , that is,

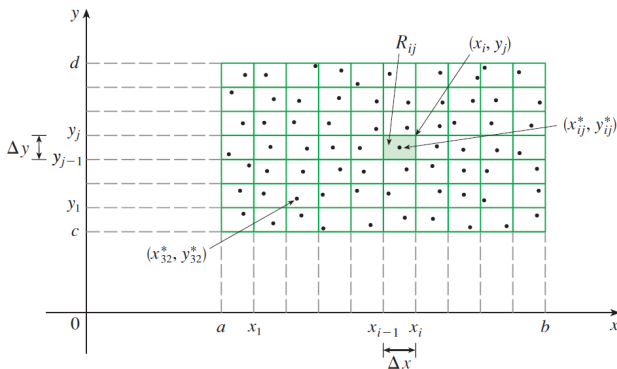
$$S = \{(x, y, z) \in \mathbb{R}^3 : 0 \leq z \leq f(x, y), (x, y) \in R\}.$$



By activity 11.1.2 (a)-(g), we partition the region R into several sub-rectangles

$$R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j] = \{(x, y) : x_{i-1} \leq x \leq x_i, y_{j-1} \leq y \leq y_j\},$$

each with area $\Delta A = \Delta x \Delta y$.



By activity 11.1.2 (a)-(g), we get an approximation to the total volume of S :

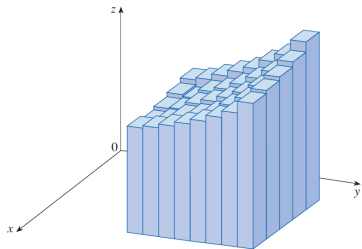
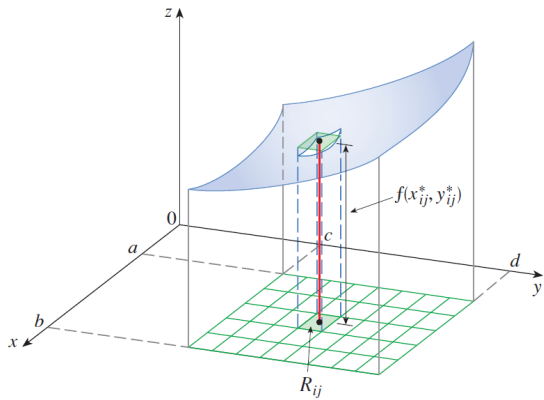
$$V \approx \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A.$$

This double sum (also called a ‘double Riemann sum’) means that for each subrectangle we evaluate f at the chosen point and multiply by the area of the subrectangle, and then we add the results.

Note

The sample point (x_{ij}^*, y_{ij}^*) can be chosen to be any point in the subrectangle R_{ij} , but if we choose it to be the upper right-hand corner of R_{ij} [namely (x_i, y_j)], then the expression of double Riemann sum becomes

$$V \approx \sum_{i=1}^m \sum_{j=1}^n f(x_i, y_j) \Delta A$$



Double integral over rectangle area

Double integral over rectangle area

Let R be a rectangular region in the xy -plane and f a continuous function over R . With terms defined as in a double Riemann sum, the **double integral of f** over R is

$$\iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A.$$

Interpretation of Double Riemann Sums and Double integrals

At the moment, there are two ways we can interpret the value of the double integral.

- Suppose that $f(x, y)$ assumes both positive and negative values on R . The product $f(x_{ij}^*, y_{ij}^*)\Delta A$ in the Riemann sum can be interpreted as a “signed” volume of a box with base area ΔA and “signed” height $f(x_{ij}^*, y_{ij}^*)$.

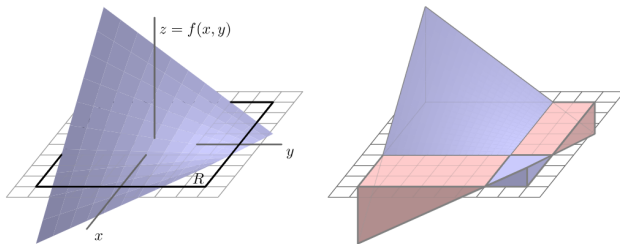


Figure 11.1.7. The integral measures signed volume.

The sum

$$\sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A$$

can then be interpreted as a sum of “signed” volumes of boxes, with a negative sign attached to those boxes whose heights are below the xy -plane.

The geometric meaning of $\iint_R f(x, y) dA$

Thus, the double integral $\iint_R f(x, y) dA$ equals the volume of the solids the graph of f bounds above the xy -plane over the rectangle R minus the volume of the solids the graph of f bounds below the xy -plane under the rectangle R .

- The average of the finitely many mn values $f(x_{ij}^*, y_{ij}^*)$ that we take in a double Riemann sum is given by

$$\text{Avg}_{mn} = \frac{1}{mn} \sum_{j=1}^n \sum_{i=1}^m f(x_{ij}^*, y_{ij}^*).$$

If we take the limit as m and n go to infinity, we obtain what we define as the average value of f over the region R . Note that

$$\Delta x = \frac{b-a}{m}, \Delta y = \frac{d-c}{n} \implies \frac{1}{mn} = \frac{\Delta x \cdot \Delta y}{(b-a)(d-c)} = \frac{\Delta A}{A(R)}.$$

Then the average value of the function f over R is given by

$$\begin{aligned} f_{\text{AVG}}(R) &= \lim_{m,n \rightarrow \infty} \frac{1}{mn} \sum_{j=1}^n \sum_{i=1}^m f(x_{ij}^*, y_{ij}^*) \\ &= \lim_{m,n \rightarrow \infty} \frac{1}{A(R)} \sum_{j=1}^n \sum_{i=1}^m f(x_{ij}^*, y_{ij}^*) \Delta A = \frac{1}{A(R)} \iint_R f(x, y) dA \end{aligned}$$

Therefore, the double integral of f over R divided by the area of R gives us the average value of the function f on R . Finally, if $f(x, y) \geq 0$ on R , we can interpret this average value of f on R as the height of the box with base R that has the same volume as the volume of the surface defined by f over R . That is,

$$f_{\text{AVG}}(R) \cdot A(R) = \iint_R f(x, y) dA$$

Activity 11.1.3. Let $f(x, y) = x + 2y$ and let $R = [0, 2] \times [1, 3]$.

- Draw a picture of R . Partition $[0, 2]$ into 2 subintervals of equal length and the interval $[1, 3]$ into two subintervals of equal length. Draw these partitions on your picture of R and label the resulting subrectangles using the labeling scheme we established in the definition of a double Riemann sum.
- For each i and j , let (x_{ij}^*, y_{ij}^*) be the midpoint of the rectangle R_{ij} . Identify the coordinates of each (x_{ij}^*, y_{ij}^*) . Draw these points on your picture of R .
- Calculate the Riemann sum

$$\sum_{j=1}^n \sum_{i=1}^m f(x_{ij}^*, y_{ij}^*) \cdot \Delta A$$

using the partitions we have described. If we let (x_{ij}^*, y_{ij}^*) be the midpoint of the rectangle R_{ij} for each i and j , then the resulting Riemann sum is called a *midpoint sum*.

- Give two interpretations for the meaning of the sum you just calculated.

Example

If

$$R = \{(x, y) : -1 \leq x \leq 1, -2 \leq y \leq 2\},$$

evaluate the integral

$$\iint_R \sqrt{1 - x^2} dA$$

Properties of Double Integrals

Properties of Double Integrals.

Let f and g be continuous functions on a rectangle

$R = \{(x, y) : a \leq x \leq b, c \leq y \leq d\}$, and let k be a constant. Then

1. $\iint_R (f(x, y) + g(x, y)) dA = \iint_R f(x, y) dA + \iint_R g(x, y) dA.$
2. $\iint_R kf(x, y) dA = k \iint_R f(x, y) dA.$
3. If $f(x, y) \geq g(x, y)$ on R , then $\iint_R f(x, y) dA \geq \iint_R g(x, y) dA.$

§11.2 Iterated Integrals

Recall that we defined the double integral of a continuous function $f = f(x, y)$ over a rectangle $R = [a, b] \times [c, d]$ as

$$\iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{j=1}^n \sum_{i=1}^m f(x_{ij}^*, y_{ij}^*) \Delta A.$$

But how to evaluate $\iint_R f(x, y) dA$?

Ideas of evaluating $\iint_R f(x, y) dA$

- We could slice the solid perpendicular to an axis and have each slice be approximately a circular disk. If we fix $x = x_i^* \in [x_{i-1}, x_i]$, the volume of the disk that perpendicular to x axis is approximately

$$\Delta x \cdot \int_c^d f(x_i^*, y) dy.$$

- And then we can add the volumes of all m disks together and let $m \rightarrow \infty$ to get the volume of the solid (this is the idea of our single variable integral). It is

$$\begin{aligned} V = \iint_R f(x, y) dA &= \lim_{m \rightarrow \infty} \sum_{i=1}^m \left[\int_c^d f(x_i^*, y) dy \right] \Delta x \\ &= \int_a^b \left[\int_c^d f(x, y) dy \right] dx \end{aligned}$$

Preview Activity 11.2.1. Let $f(x, y) = 25 - x^2 - y^2$ on the rectangular domain $R = [-3, 3] \times [-4, 4]$.

As with partial derivatives, we may treat one of the variables in f as constant and think of the resulting function as a function of a single variable. Now we investigate what happens if we integrate instead of differentiate.

- a. Choose a fixed value of x in the interior of $[-3, 3]$. Let

$$A(x) = \int_{-4}^4 f(x, y) dy.$$

What is the geometric meaning of the value of $A(x)$ relative to the surface defined by f . (Hint: Think about the trace determined by the fixed value of x , and consider how $A(x)$ is related to the image at left in [Figure 11.2.2.](#))

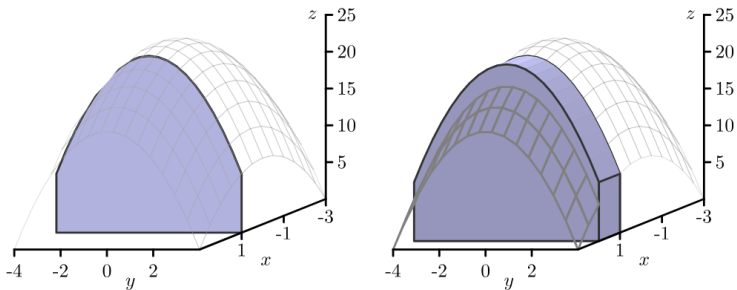


Figure 11.2.2. Left: A cross section with fixed x . Right: A cross section with fixed x and Δx .

- b. For a fixed value of x , say x_i^* , what is the geometric meaning of $A(x_i^*) \Delta x$? (Hint: Consider how $A(x_i^*) \Delta x$ is related to the image at right in [Figure 11.2.2.](#))
- c. Since f is continuous on R , we can define the function $A = A(x)$ at every value of x in $[-3, 3]$. Now think about subdividing the x -interval $[-3, 3]$ into m subintervals, and choosing a value x_i^* in each of those subintervals. What will be the meaning of the sum $\sum_{i=1}^m A(x_i^*) \Delta x$?
- d. Explain why $\int_{-3}^3 A(x) dx$ will determine the exact value of the volume under the surface $z = f(x, y)$ over the rectangle R .

Iterated Integrals

Let f be a continuous function on $R = [a, b] \times [c, d]$.

- If we fix x , the cross sectional area in the y direction of the solid is

$$A(x) = \int_c^d f(x, y) dy.$$

- The integral

$$\int_a^b A(x) dx \approx \sum_{i=1}^m A(x_i^*) \Delta x$$

is the volume of the solid as each term $A(x_i^*) \Delta x$ represents an approximation of a fixed cross sectional slice of the surface in the y direction with a fixed width of Δx . By evaluating the total volume we sum all slices up and let $\Delta x \rightarrow 0 (m \rightarrow \infty)$.

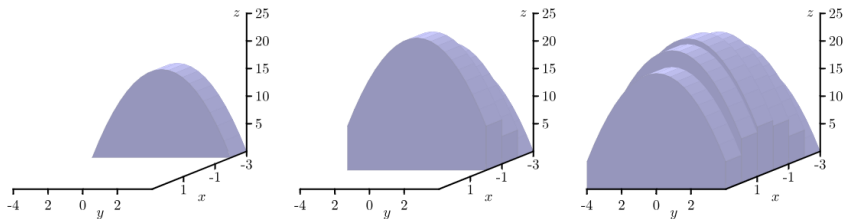


Figure 11.2.3. Summing volumes of cross section slices.

Therefore, we have

$$\begin{aligned} \iint_R f(x, y) dA &= \lim_{m \rightarrow \infty} \sum_{i=1}^m A(x_i^*) \Delta x = \int_a^b A(x) dx \\ &= \int_a^b \left(\int_c^d f(x, y) dy \right) dx. \end{aligned}$$

The nested integral

$$\int_a^b \left(\int_c^d f(x, y) dy \right) dx = \int_a^b \int_c^d f(x, y) dy dx$$

is called an *iterated integral*.

Similarly, if we fix y and sum all slices that perpendicular to y -axis up, we get the same volume. That is,

$$\iint_R f(x, y) dA = \int_c^d \int_a^b f(x, y) dx dy$$

Fubini's Theorem

The fact that integrating in either order results in the same value is known as Fubini's Theorem.

Fubini's Theorem

If $f(x, y)$ is a continuous function on $R = [a, b] \times [c, d]$, then

$$\iint_R f(x, y) dA = \int_c^d \int_a^b f(x, y) dx dy = \int_a^b \int_c^d f(x, y) dy dx$$

Activity 11.2.2. Let $f(x, y) = 25 - x^2 - y^2$ on the rectangular domain $R = [-3, 3] \times [-4, 4]$.

- a. Viewing x as a fixed constant, use the Fundamental Theorem of Calculus to evaluate the integral

$$A(x) = \int_{-4}^4 f(x, y) dy.$$

Note that you will be integrating with respect to y , and holding x constant. Your result should be a function of x only.

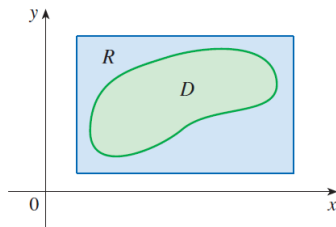
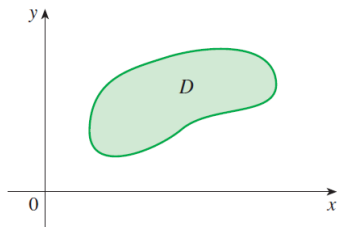
- b. Next, use your result from (a) along with the Fundamental Theorem of Calculus to determine the value of $\int_{-3}^3 A(x) dx$.
- c. What is the value of $\iint_R f(x, y) dA$? What are two different ways we may interpret the meaning of this value?

Activity 11.2.3. Let $f(x, y) = x + y^2$ on the rectangle $R = [0, 2] \times [0, 3]$.

- a. Evaluate $\iint_R f(x, y) \, dA$ using an iterated integral. Choose an order for integration by deciding whether you want to integrate first with respect to x or y .
- b. Evaluate $\iint_R f(x, y) \, dA$ using the iterated integral whose order of integration is the opposite of the order you chose in (a).

§11.3 Double Integrals over general regions

For double integrals, we want to be able to integrate a function not just over rectangles but also over regions of more general shape. Suppose that D is a bounded region, which means that D can be enclosed in a rectangular region R .



In order to integrate a function f over D we define a new function F with domain R by

$$F(x, y) = \begin{cases} f(x, y), & \text{if } (x, y) \in D \\ 0, & \text{if } (x, y) \notin D. \end{cases}$$

If F is integrable over R , then we define the double integral of f over D by

$$\boxed{\iint_D f(x, y) dA = \iint_R F(x, y) dA.}$$

Now we have defined the double integral over D , but how to compute $\iint_D f(x, y) dA$?

Preview Activity 11.3.1. A tetrahedron is a three-dimensional figure with four faces, each of which is a triangle. A picture of the tetrahedron T with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ is shown at left in [Figure 11.3.1](#). If we place one vertex at the origin and let vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ be determined by the edges of the tetrahedron that have one end at the origin, then a formula that tells us the volume V of the tetrahedron is

$$V = \frac{1}{6} |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|. \quad (11.3.1)$$

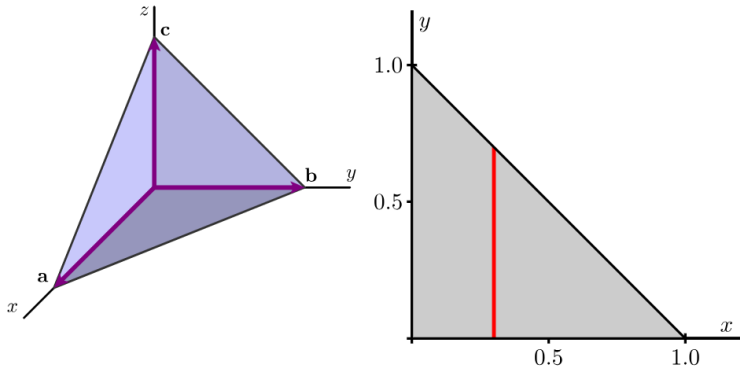


Figure 11.3.1. Left: The tetrahedron T . Right: Projecting T onto the xy -plane.

- a. Use the formula [\(11.3.1\)](#) to find the volume of the tetrahedron T .
- b. Instead of memorizing or looking up the formula for the volume of a tetrahedron, we can use a double integral to calculate the volume of the tetrahedron T . To see how, notice that the top face of the tetrahedron T is the plane whose equation is

$$z = 1 - (x + y).$$

Provided that we can use an iterated integral on a non-rectangular region, the volume of the tetrahedron will be given by an iterated integral of the form

$$\int_{x=?}^{x=?} \int_{y=?}^{y=?} 1 - (x + y) \, dy \, dx.$$

The issue that is new here is how we find the limits on the integrals; note that the outer integral's limits are in x , while the inner ones are in y , since we have chosen $dA = dy dx$. To see the domain over which we need to integrate, think of standing way above the tetrahedron looking straight down on it, which means we are projecting the entire tetrahedron onto the xy -plane. The resulting domain is the triangular region shown at right in [Figure 11.3.1](#). Explain why we can represent the triangular region with the inequalities

$$0 \leq y \leq 1 - x \quad \text{and} \quad 0 \leq x \leq 1.$$

(Hint: Consider the cross sectional slice shown at right in [Figure 11.3.1](#).)

- c. Explain why it makes sense to now write the volume integral in the form

$$\int_{x=?}^{x=?} \int_{y=?}^{y=?} 1 - (x + y) dy dx = \int_{x=0}^{x=1} \int_{y=0}^{y=1-x} 1 - (x + y) dy dx.$$

- d. Use the Fundamental Theorem of Calculus to evaluate the iterated integral

$$\int_{x=0}^{x=1} \int_{y=0}^{y=1-x} 1 - (x + y) dy dx$$

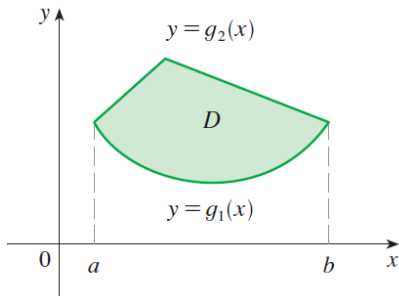
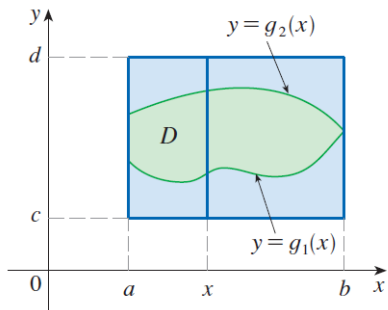
and compare to your result from part (a). (As with iterated integrals over rectangular regions, start with the inner integral.)

Double Integrals over General Regions

Just as with double integrals over rectangles, a double integral over a domain D can be evaluated as an iterated integral.

- Type I region: If the region D can be described by the inequalities $g_1(x) \leq y \leq g_2(x)$ and $a \leq x \leq b$, where $g_1 = g_1(x)$ and $g_2 = g_2(x)$ are functions of only x , then

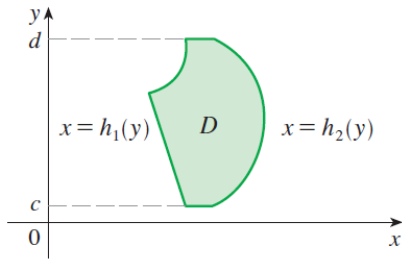
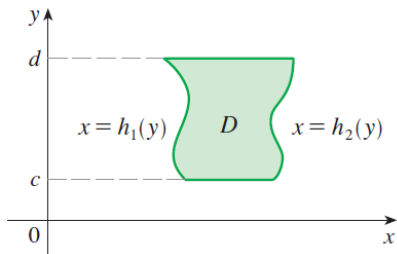
$$\iint_D f(x, y) dA = \int_{x=a}^{x=b} \int_{y=g_1(x)}^{y=g_2(x)} f(x, y) dy dx$$



Note: For a type I region, the functions g_1 and g_2 must be continuous but they do not need to be defined by a single formula.

- Type II region: If the region D is described by the inequalities $h_1(y) \leq x \leq h_2(y)$ and $c \leq y \leq d$, where $h_1 = h_1(y)$ and $h_2 = h_2(y)$ are functions of only y , we have

$$\iint_D f(x, y) dA = \int_{y=c}^{y=d} \int_{x=h_1(y)}^{x=h_2(y)} f(x, y) dx dy.$$



Note: For a type II region, the functions h_1 and h_2 must be continuous but they do not need to be defined by a single formula.

The structure of an iterated integral is of particular note:

In an iterated double integral:

- the limits on the outer integral must be constants;
- the limits on the inner integral must be constants or in terms of only the remaining variable - that is, if the inner integral is with respect to y , then its limits may only involve x and constants, and vice versa.

A detailed example

Example 11.3.3. Let $f(x, y) = x^2y$ be defined on the triangle D with vertices $(0, 0)$, $(2, 0)$, and $(2, 3)$ as shown at left in [Figure 11.3.4](#).

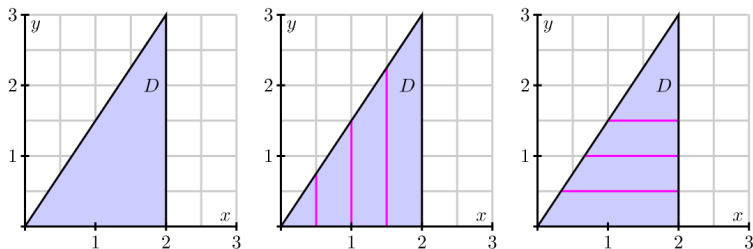


Figure 11.3.4. A triangular domain and slices in the y and x directions.

To evaluate $\iint_D f(x, y) dA$, we must first describe the region D in terms of the variables x and y . We take two approaches.

Approach 1: Integrate first with respect to y .

In this case we choose to evaluate the double integral as an iterated integral in the form

$$\iint_D f(x, y) dA = \int_{x=a}^{x=b} \int_{y=g_1(x)}^{y=g_2(x)} f(x, y) dy dx,$$

and therefore we need to describe D in terms of inequalities

$$g_1(x) \leq y \leq g_2(x) \quad \text{and} \quad a \leq x \leq b.$$

Approach 1 ct'd

Since we are integrating with respect to y first, the iterated integral has the form

$$\iint_D x^2 y dA = \int_{x=a}^{x=b} A(x) dx,$$

where $A(x)$ is a cross sectional area in the y direction. So we are slicing the domain perpendicular to the x -axis and want to understand what a cross sectional area of the overall solid will look like.

- On a slice with fixed x value, the y values are bounded below by 0 and above by the y coordinate on the hypotenuse of the right triangle. Thus, $g_1(x) = 0$;
- to find $y = g_2(x)$, we need to write the hypotenuse as a function of x . The hypotenuse connects the points $(0, 0)$ and $(2, 3)$ and hence has equation $y = \frac{3}{2}x$. This gives the upper bound on y as $g_2(x) = \frac{3}{2}x$.

Approach 1 ct'd

The leftmost vertical cross section is at $x = 0$ and the rightmost one is at $x = 2$, so we have $a = 0$ and $b = 2$. Therefore,

$$\iint_D x^2 y dA = \int_{x=0}^{x=2} \int_{y=0}^{y=\frac{3}{2}x} x^2 y dy dx.$$

We evaluate the iterated integral by applying the Fundamental Theorem of Calculus first to the inner integral, and then to the outer one, and find that

$$\begin{aligned} \int_{x=0}^{x=2} \int_{y=0}^{y=\frac{3}{2}x} x^2 y dy dx &= \int_0^2 \left[x^2 \cdot \frac{y^2}{2} \right] \Big|_{y=0}^{y=\frac{3}{2}x} dx \\ &= \int_0^2 \frac{9}{8} x^4 dx = \frac{9}{40} x^5 \Big|_{x=0}^{x=2} = \frac{36}{5}. \end{aligned}$$

Approach 2: Integrate first with respect to x .

In this case, we choose to evaluate the double integral as an iterated integral in the form

$$\iint_D f(x, y) dA = \int_{y=c}^{y=d} \int_{x=h_1(y)}^{x=h_2(y)} f(x, y) dx dy$$

and thus need to describe D in terms of inequalities

$$h_1(y) \leq x \leq h_2(y) \quad \text{and} \quad c \leq y \leq d.$$

Approach 2 ct'd

Since we are integrating with respect to x first, the iterated integral has the form

$$\iint_D x^2 y dA = \int_{y=c}^{y=d} A(y) dy,$$

where $A(y)$ is a cross sectional area in the x direction. So we are slicing the domain perpendicular to the y -axis and want to understand what a cross sectional area of the overall solid will look like.

- On a slice with fixed y value, the x values are bounded below by the x coordinate on the hypotenuse of the right triangle and above by 2. Thus, $h_2(y) = 2$;
- to find $x = h_1(y)$, we need to write the hypotenuse as a function of y . The hypotenuse connects the points $(0, 0)$ and $(2, 3)$ and hence has equation $y = \frac{3}{2}x \implies x = \frac{2}{3}y$. This gives the lower bound on x as $h_1(y) = \frac{2}{3}y$.

Approach 2 ct'd

The lowest horizontal cross section is at $y = 0$ and the uppermost one is at $y = 3$, so we have $c = 0$ and $d = 3$. Therefore,

$$\iint_D x^2 y dA = \int_{y=0}^{y=3} \int_{x=\frac{2}{3}y}^{x=2} x^2 y dx dy.$$

We evaluate the iterated integral by applying the Fundamental Theorem of Calculus first to the inner integral, and then to the outer one, and find that

$$\begin{aligned} \int_{y=0}^{y=3} \int_{x=\frac{2}{3}y}^{x=2} x^2 y dx dy &= \int_0^3 \left[y \cdot \frac{x^3}{3} \right] \Big|_{x=\frac{2}{3}y}^{x=2} dy \\ &= \int_0^3 \left[\frac{8}{3}y - \frac{8}{81}y^4 \right] dy \\ &= \left[\frac{4}{3}y^2 - \frac{8}{405}y^5 \right] \Big|_{y=0}^{y=3} = 12 - \frac{24}{5} = \frac{36}{5}. \end{aligned}$$

The meaning of a double integral over a non-rectangular region, D , parallels the meaning over a rectangular region. In particular,

- $\iint_D f(x, y) dA$ tells us the volume of the solids the graph of f bounds above the xy -plane over the closed, bounded region D minus the volume of the solids the graph of f bounds below the xy -plane under the region D ;
- $\frac{1}{A(D)} \iint_D f(x, y) dA$, where $A(D)$ is the area of D tells us the average value of the function f on D . If $f(x, y) \geq 0$ on D , we can interpret this average value of f on D as the height of the solid with base D and constant cross-sectional area D that has the same volume as the volume of the surface defined by f over D .

Activity 11.3.2. Consider the double integral $\iint_D (4 - x - 2y) \, dA$, where D is the triangular region with vertices $(0,0)$, $(4,0)$, and $(0,2)$.

- Write the given integral as an iterated integral of the form $\iint_D (4 - x - 2y) \, dy \, dx$. Draw a labeled picture of D with relevant cross sections.
- Write the given integral as an iterated integral of the form $\iint_D (4 - x - 2y) \, dx \, dy$. Draw a labeled picture of D with relevant cross sections.
- Evaluate the two iterated integrals from (a) and (b), and verify that they produce the same value. Give at least one interpretation of the meaning of your result.

Activity 11.3.3. Consider the iterated integral

$$\int_{x=0}^{x=1} \int_{y=x}^{y=\sqrt{x}} (4x + 10y) dy dx.$$

- a. Sketch the region of integration, D , for which

$$\iint_D (4x + 10y) dA = \int_{x=0}^{x=1} \int_{y=x}^{y=\sqrt{x}} (4x + 10y) dy dx.$$

- b. Determine the equivalent iterated integral that results from integrating in the opposite order ($dx dy$, instead of $dy dx$). That is, determine the limits of integration for which

$$\iint_D (4x + 10y) dA = \int_{y=?}^{y=?} \int_{x=?}^{x=?} (4x + 10y) dx dy.$$

- c. Evaluate one of the two iterated integrals above. Explain what the value you obtained tells you.
- d. Set up and evaluate a single definite integral to determine the exact area of D , $A(D)$.
- e. Determine the exact average value of $f(x, y) = 4x + 10y$ over D .

Activity 11.3.4. Consider the iterated integral $\int_{x=0}^{x=4} \int_{y=x/2}^{y=2} e^{y^2} dy dx$.

- Explain why we cannot find a simple antiderivative for e^{y^2} with respect to y , and thus are unable to evaluate $\int_{x=0}^{x=4} \int_{y=x/2}^{y=2} e^{y^2} dy dx$ in the indicated order using the Fundamental Theorem of Calculus.
- Given that $\iint_D e^{y^2} dA = \int_{x=0}^{x=4} \int_{y=x/2}^{y=2} e^{y^2} dy dx$, sketch the region of integration, D .
- Rewrite the given iterated integral in the opposite order, using $dA = dx dy$. (Hint: You may need more than one integral.)
- Use the Fundamental Theorem of Calculus to evaluate the iterated integral you developed in (c). Write one sentence to explain the meaning of the value you found.
- What is the important lesson this activity offers regarding the order in which we set up an iterated integral?

§11.4 Double Integrals in Polar Coordinates

While we have naturally defined double integrals in the rectangular coordinate system, starting with domains that are rectangular regions, there are many of these integrals that are difficult, especially when the domain D is a circle or ellipse.

Example

Consider the domain

$$D = \{(x, y) : 0 \leq x^2 + y^2 \leq 1\}$$

and $f(x, y) = e^{-x^2-y^2}$. To integrate f over D , we would use the iterated integral

$$\iint_D f(x, y) dA = \int_{x=-1}^{x=1} \int_{y=-\sqrt{1-x^2}}^{y=\sqrt{1-x^2}} e^{-x^2-y^2} dy dx.$$

For this particular integral, regardless of the order of integration, we are unable to find an antiderivative of the integrand. Therefore, we need to translate to another coordinate system in which the involved integral is simpler. When the domain D is a circle or ellipse, the polar coordinates would be very useful. We will talk about the polar coordinate system and the double integration in this system.

Preview Activity 11.5.1. The coordinates of a point determine its location. In particular, the rectangular coordinates of a point P are given by an ordered pair (x, y) , where x is the (signed) distance the point lies from the y -axis to P and y is the (signed) distance the point lies from the x -axis to P . In polar coordinates, we locate the point by considering the distance the point lies from the origin, $O = (0, 0)$, and the angle the line segment from the origin to P forms with the positive x -axis.

a. Determine the rectangular coordinates of the following points:

- (a) The point P that lies 1 unit from the origin on the positive x -axis.
- (b) The point Q that lies 2 units from the origin and such that $\overline{OQ}$ makes an angle of $\frac{\pi}{2}$ with the positive x -axis.
- (c) The point R that lies 3 units from the origin such that $\overline{OR}$ makes an angle of $\frac{2\pi}{3}$ with the positive x -axis.

- b. Part (a) indicates that the two pieces of information completely determine the location of a point: either the traditional (x, y) coordinates, or alternately, the distance r from the point to the origin along with the angle θ that the line through the origin and the point makes with the positive x -axis. We write " (r, θ) " to denote the point's location in its polar coordinate representation. Find polar coordinates for the points with the given rectangular coordinates.

i. $(0, -1)$ ii. $(-2, 0)$ iii. $(-1, 1)$

- c. For each of the following points whose coordinates are given in polar form, determine the rectangular coordinates of the point.

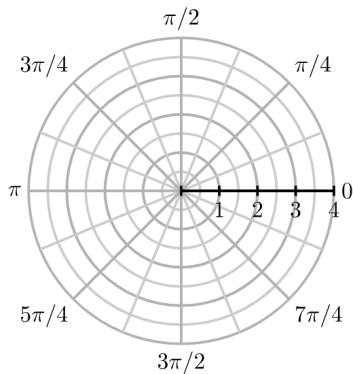
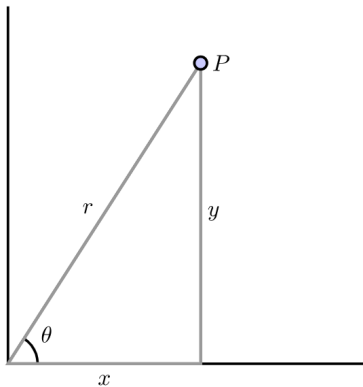
i. $(5, \frac{\pi}{4})$ ii. $(2, \frac{5\pi}{6})$ iii. $(\sqrt{3}, \frac{5\pi}{3})$

Polar Coordinates

- The rectangular coordinate system is best suited for graphs and regions that are naturally considered over a rectangular grid.
- The polar coordinate system is an alternative that offers good options for functions and domains that have more circular characteristics.

A point $P(x, y)$ in rectangular coordinate system can also be described with polar coordinates (r, θ) , where

- r is the distance from P to the origin,
- θ is the angle formed by the line segment $\overline{OP}$ and the positive x -axis.



Converting between rectangular and polar coordinates.

- *Converting from rectangular to polar..*

If we are given the rectangular coordinates (x, y) of a point P , then the polar coordinates (r, θ) of P satisfy

$$r = \sqrt{x^2 + y^2} \quad \text{and} \quad \tan(\theta) = \frac{y}{x}, \text{ assuming } x \neq 0.$$

- *Converting from polar to rectangular..*

If we are given the polar coordinates (r, θ) of a point P , then the rectangular coordinates (x, y) of P satisfy

$$x = r \cos(\theta) \quad \text{and} \quad y = r \sin(\theta).$$

Note

More care has to be paid when using polar coordinates than rectangular coordinates.

- The angle θ in the polar coordinates of a point is not unique. We could replace θ with $\theta + 2\pi$ and still be at the same terminal point.
- In addition, the sign of $\tan(\theta)$ does not uniquely determine the quadrant in which θ lies, so we have to determine the value of θ from the location of the point.

Activity 11.5.2. Most polar graphing devices can plot curves in polar coordinates of the form $r = f(\theta)$. Use such a device to complete this activity.

- Before plotting the polar curve $r = 1$ (where θ can have any value), think about what shape it should have, in light of how r is connected to x and y . Then use appropriate technology to draw the graph and test your intuition.
- The equation $\theta = 1$ does not define r as a function of θ , so we can't graph this equation on many polar plotters. What do you think the graph of the polar curve $\theta = 1$ looks like? Why?
- Before plotting the polar curve $r = \theta$, what do you think the graph looks like? Why? Use technology to plot the curve and compare your intuition.
- What does the region defined by $1 \leq r \leq 3$ (where θ can have any value) look like? (Hint: Compare to your response from part (a).)
- What does the region defined by $1 \leq r \leq 3$ and $\pi/4 \leq \theta \leq \pi/2$ look like?
- Consider the curve $r = \sin(\theta)$. For some values of θ we will have $r < 0$. In these situations, we plot the point (r, θ) as $(|r|, \theta + \pi)$ (in other words, when $r < 0$, we reflect the point through the origin). With that in mind, what do you think the graph of $r = \sin(\theta)$ looks like? Plot this curve using technology and compare to your intuition.

Integrating in Polar Coordinates

In rectangular coordinate, $dA = dx dy$, so we know how to evaluate a double integral $\iint_D f(x, y) dA$ as an iterated integral of the form

$$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx.$$

Question

What is dA in polar coordinate system? Is $dA = dr d\theta$ correct?

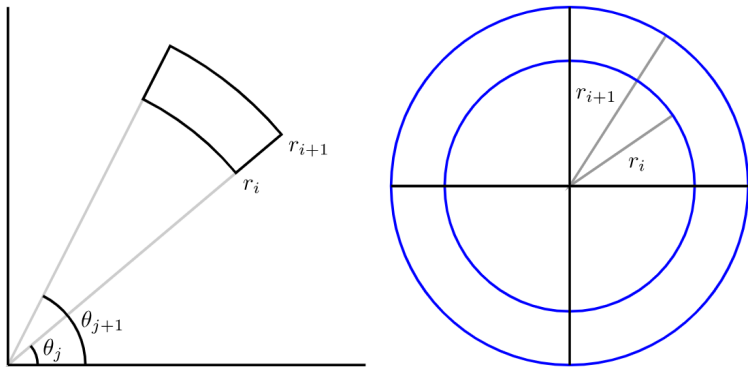


Figure 11.5.2. Left: A polar rectangle. Right: An annulus.

Activity 11.5.3. Consider a polar rectangle R , with r between r_i and r_{i+1} and θ between θ_j and θ_{j+1} as shown at left in [Figure 11.5.2](#). Let $\Delta r = r_{i+1} - r_i$ and $\Delta \theta = \theta_{j+1} - \theta_j$. Let ΔA be the area of this region.

- a. Explain why the area ΔA in polar coordinates is not $\Delta r \Delta \theta$.
- b. Now find ΔA by the following steps:
 - i. Find the area of the annulus (the washer-like region) between r_i and r_{i+1} , as shown at right in [Figure 11.5.2](#). This area will be in terms of r_i and r_{i+1} .
 - ii. Observe that the region R is only a portion of the annulus, so the area ΔA of R is only a fraction of the area of the annulus. For instance, if $\theta_{i+1} - \theta_i$ were $\frac{\pi}{4}$, then the resulting wedge would be

$$\frac{\frac{\pi}{4}}{2\pi} = \frac{1}{8}$$

of the entire annulus. In this more general context, using the wedge between the two noted angles, what fraction of the area of the annulus is the area ΔA ?

- iii. Write an expression for ΔA in terms of r_i , r_{i+1} , θ_j , and θ_{j+1} .
- iv. Finally, write the area ΔA in terms of r_i , r_{i+1} , Δr , and $\Delta \theta$, where each quantity appears only once in the expression. (Hint: Think about how to factor a difference of squares.)
- c. As we take the limit as Δr and $\Delta \theta$ go to 0, Δr becomes dr , $\Delta \theta$ becomes $d\theta$, and ΔA becomes dA , the area element. Using your work in (iv), write dA in terms of r , dr , and $d\theta$.

From the result of Activity 11.5.3,

- when we convert an integral from rectangular coordinates to polar coordinates, we must not only convert x and y to being in terms of r and θ , but we also have to change the area element to $dA = r dr d\theta$ in polar coordinates.
- the reason the additional factor of r in the polar area element is due to the fact that in polar coordinates, the cross sectional area element increases as r increases, while the cross sectional area element in rectangular coordinates is constant.

Double integrals in polar coordinates.

The double integral $\iint_D f(x, y) dA$ in rectangular coordinates can be converted to a double integral in polar coordinates as

$$\iint_D f(r \cos(\theta), r \sin(\theta)) r dr d\theta.$$

A detailed example

Example

Let $f(x, y) = e^{-x^2-y^2}$ on the disk $D = \{(x, y) : x^2 + y^2 \leq 1\}$. Evaluate $\iint_D f(x, y) dA$.

In rectangular coordinates the integral is really hard to evaluate. Since $r^2 = x^2 + y^2$ and the region D is circular, it is natural to wonder whether converting to polar coordinates will allow us to evaluate the new integral. To do so, we

- replace x with $r \cos(\theta)$
- replace y with $r \sin(\theta)$
- replace $dydx$ with $rdrd\theta$

to obtain

$$\iint_D f(x, y) dA = \iint_D e^{-r^2} r dr d\theta.$$

Example ct'd

The disc D is described in polar coordinates by the following set

$$D = \{(r, \theta) : 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi\}.$$

Therefore, it follows that

$$\iint_D f(x, y) dA = \iint_D e^{-r^2} r dr d\theta = \int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=1} e^{-r^2} r dr d\theta.$$

Example ct'd

We can evaluate the resulting iterated polar integral as follows:

$$\begin{aligned}\int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=1} e^{-r^2} r dr d\theta &= \int_0^{2\pi} \left(-\frac{1}{2} e^{-r^2} \Big|_{r=0}^{r=1} \right) d\theta \\ &= \frac{1}{2} \int_0^{2\pi} (1 - e^{-1}) d\theta \\ &= \frac{1}{2} (1 - e^{-1}) [\theta] \Big|_{\theta=0}^{\theta=2\pi} = \pi(1 - e^{-1}).\end{aligned}$$

Activity 11.5.4. Let $f(x, y) = x + y$ and $D = \{(x, y) : x^2 + y^2 \leq 4\}$.

- Sketch the region D and then write the double integral of f over D as an iterated integral in rectangular coordinates.
- Write the double integral of f over D as an iterated integral in polar coordinates.
- Evaluate one of the iterated integrals. Why is the final value you found not surprising?

Activity 11.5.5. Consider the circle given by $x^2 + (y - 1)^2 = 1$ as shown in [Figure 11.5.4](#).

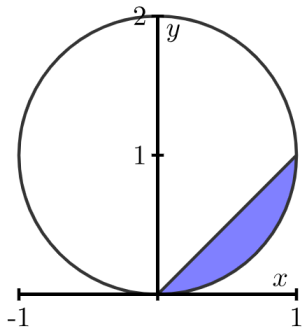


Figure 11.5.4. The graphs of $y = x$ and $x^2 + (y - 1)^2 = 1$, for use in [Activity 11.5.5](#).

- a. Determine a polar curve in the form $r = f(\theta)$ that traces out the circle $x^2 + (y - 1)^2 = 1$. (Hint: Recall that a circle centered at the origin of radius r can be described by the equations $x = r \cos(\theta)$ and $y = r \sin(\theta)$.)
- b. Find the exact average value of $g(x, y) = \sqrt{x^2 + y^2}$ over the interior of the circle $x^2 + (y - 1)^2 = 1$.
- c. Find the volume under the surface $h(x, y) = x$ over the region D , where D is the region bounded above by the line $y = x$ and below by the circle (this is the shaded region in [Figure 11.5.4](#)).
- d. Explain why in both (b) and (c) it is advantageous to use polar coordinates.